

Quantum algorithms for nonlinear dynamics

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AGENDA

0 1 Motivation

Potential quantum speedups

0 2 Linear algebra on quantum computers

Review of the existing results

0 3 Quantum algorithms for nonlinear dynamics

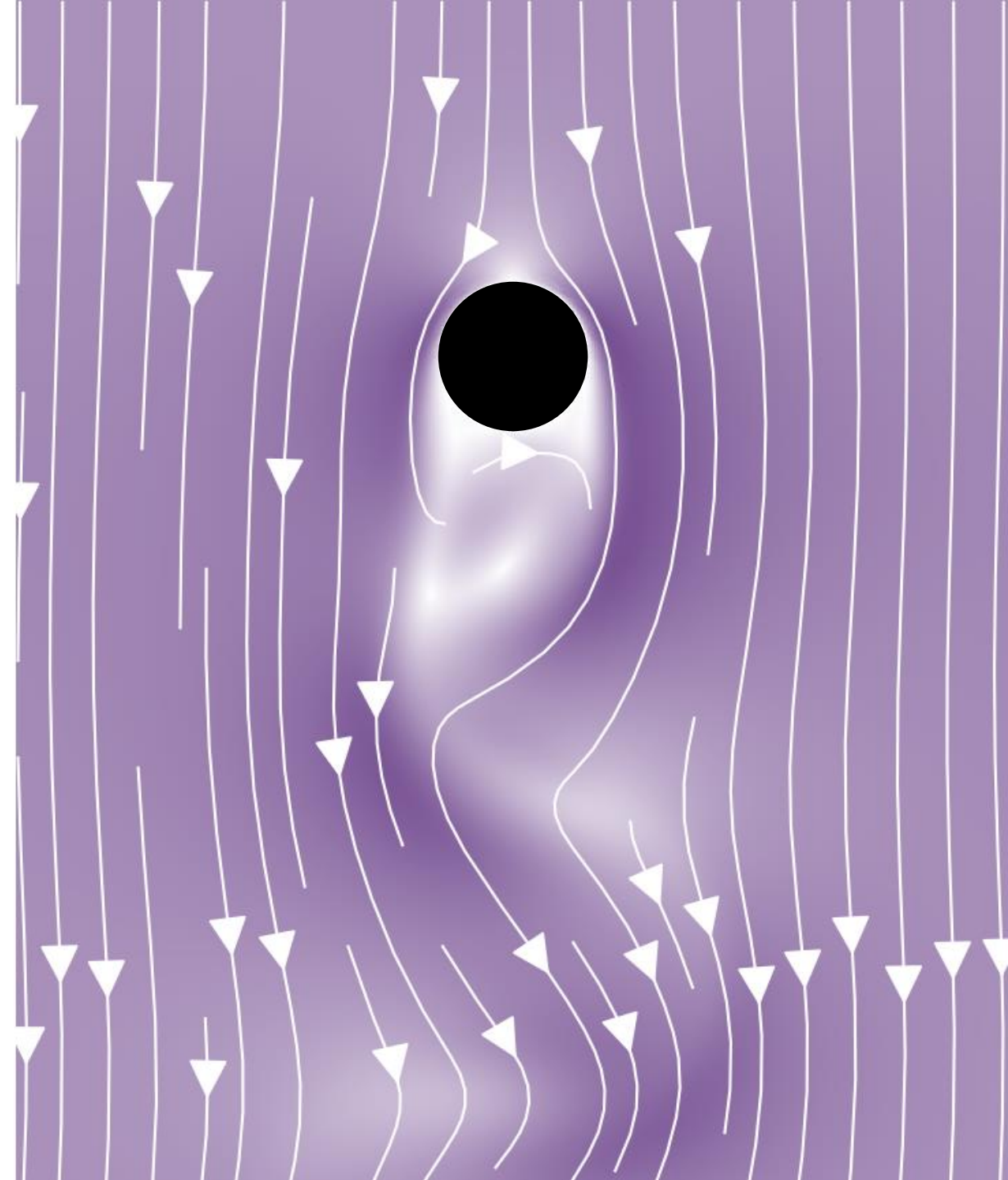
Novel results from PsiQuantum

0 4 Application to fluid dynamics

PsiQuantum testing industrial relevance

0 5 Outlook

Path for future research



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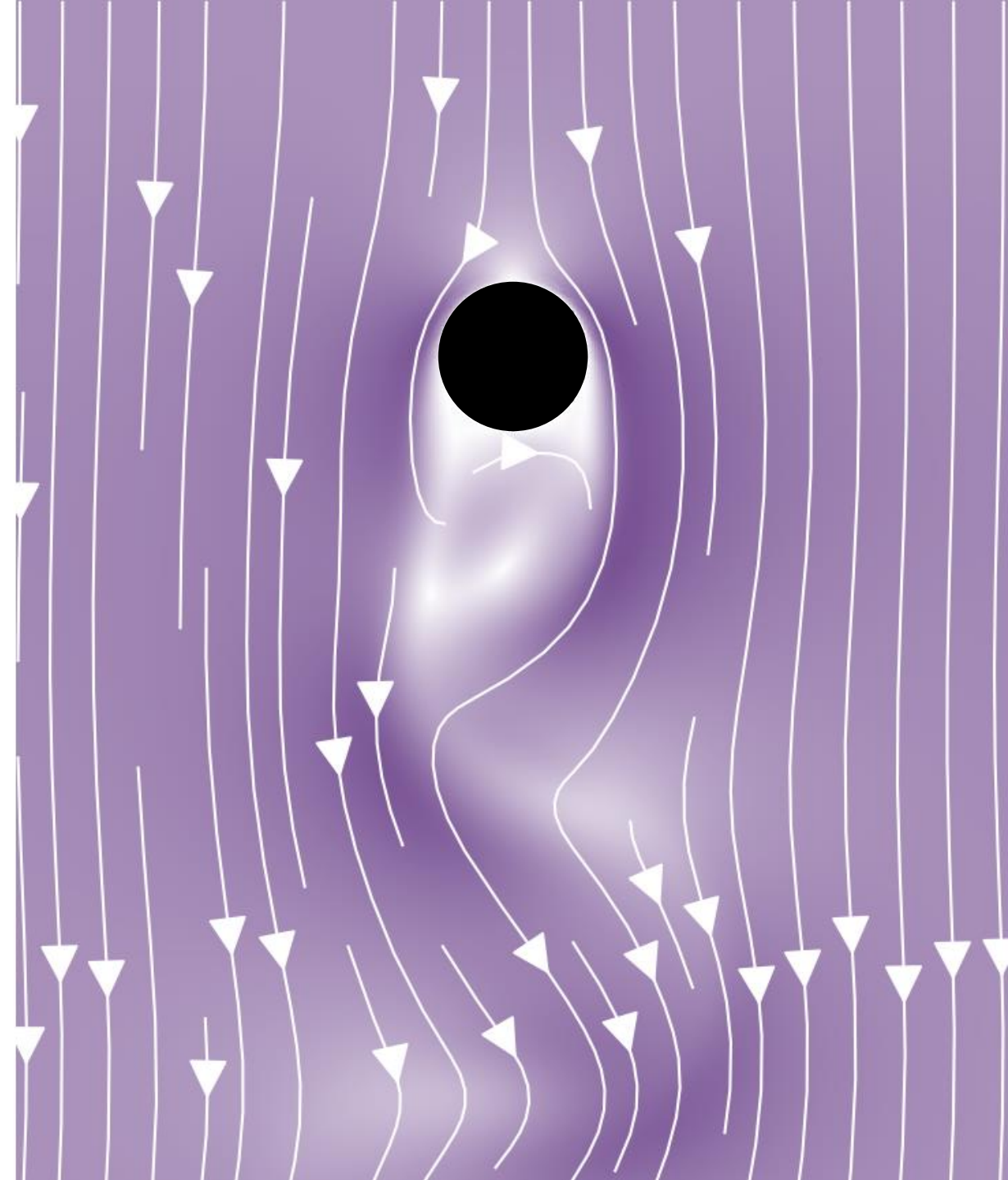
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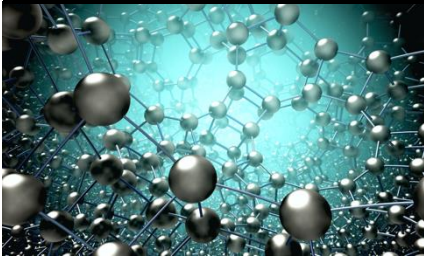


MOTIVATION

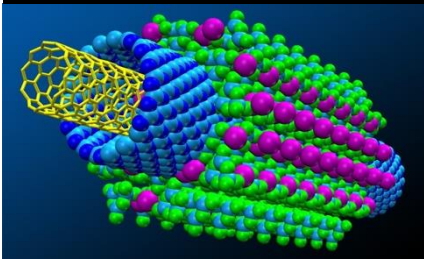
Quantum computing use-cases

Quantum Chemistry

Molecular Properties

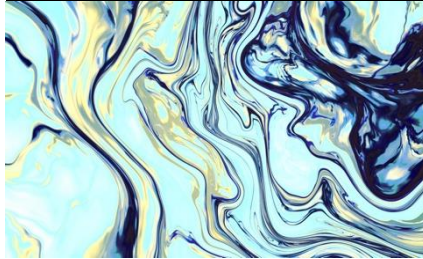


Materials Screening



Quantum PDE Solvers

Fluid Dynamics



Plasma Physics

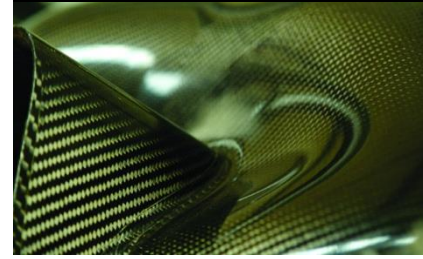


Quantum Optimization

Logistics & Routing



Structure Design



Quantum ML

Predictive Maintenance



Neural Networks



Exponential

High-Degree Polynomial

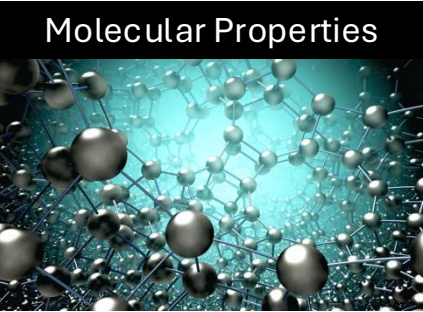
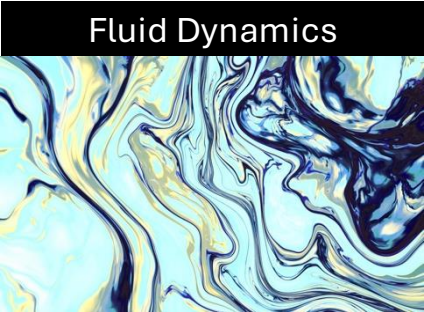
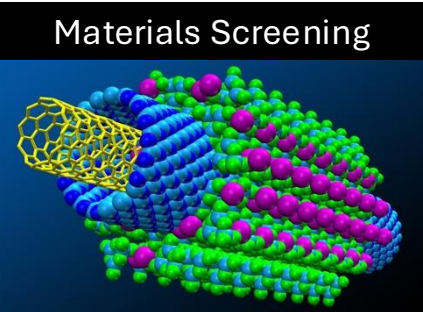
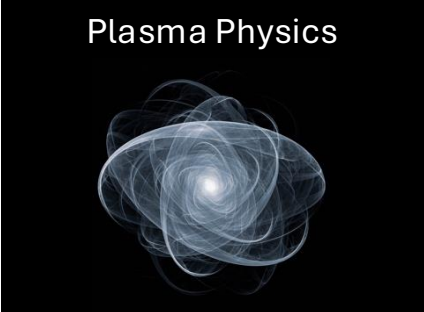
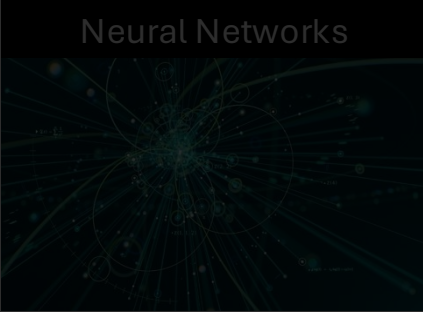
Low-Degree Polynomial

TBD?

Expected Quantum Speed Up

MOTIVATION

Quantum computing use-cases

Quantum Chemistry	Quantum PDE Solvers	Quantum Optimization	Quantum ML
Molecular Properties 	Fluid Dynamics 	Logistics & Routing 	Predictive Maintenance 
Materials Screening 	Plasma Physics 	Structure Design 	Neural Networks 
Exponential	High-Degree Polynomial	Low-Degree Polynomial	TBD?

Expected Quantum Speed Up

MOTIVATION

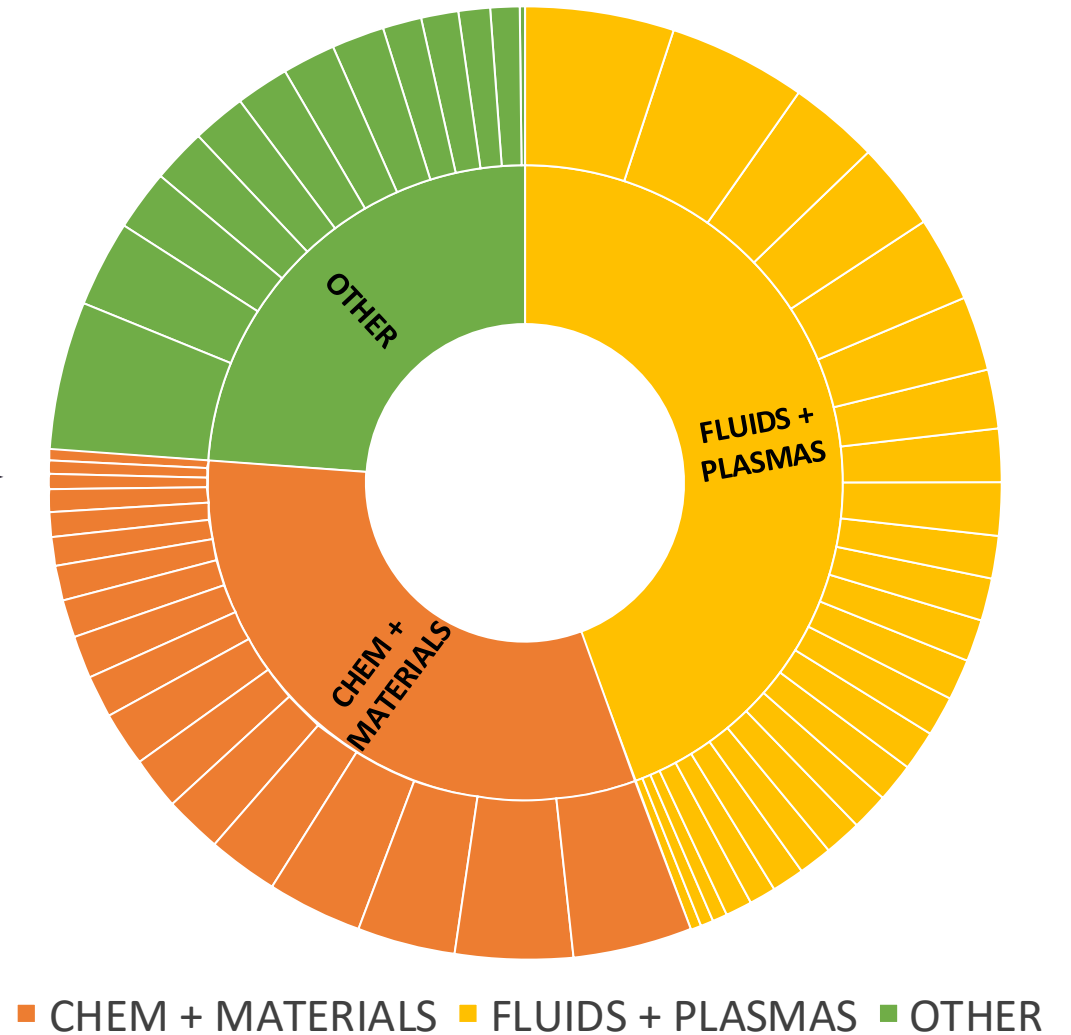
Importance of nonlinear problems

A substantial fraction of today's high-performance supercomputing is devoted to the resolution of these problems

DOE INCITE 2023 projects by core hours:
>55M core-hours allocated over 56 projects

PsiQuantum's goal:

Provable quantum advantage on a high-impact classical dynamical system, with detailed costing and no caveats.



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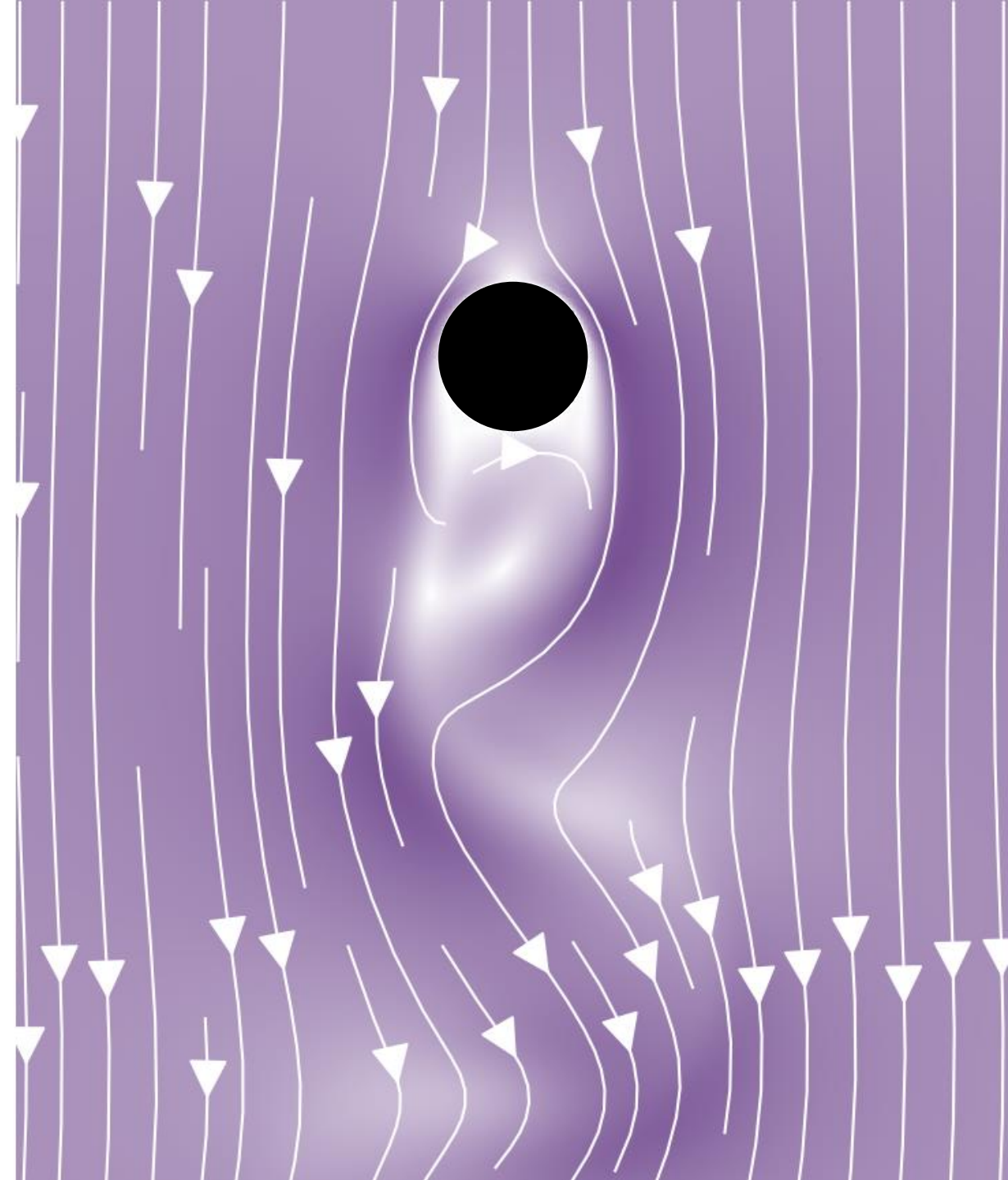
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LINEAR ALGEBRA ON QUANTUM COMPUTERS

The origin of quantum speedup

Time evolution of a quantum system is described by the Schrödinger equation:

$$i \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle \xrightarrow{\text{solve}} |\psi(t)\rangle = U(t) |\psi(0)\rangle \quad \text{with a unitary } U(t) = e^{-iHt}$$

Problem: $|\psi(t)\rangle$ lives in a d -dimensional Hilbert space, with d exponential in the system size N .

“...because nature isn't classical, dammit, and if you want to make a simulation of nature, you'd better make it quantum mechanical, and by golly it's a wonderful problem, because it doesn't look so easy.”

R. Feynman in *Simulating Physics with Computers*

“Quantum computing is powerful because unitary operators describing the time-evolution of a quantum system have exponential size in terms of the number of qubits present in the system.”

arXiv:1806.01838

Unitary block-encodings

Question: Can we embed more general problems into the Schrödinger evolution and obtain a quantum speedup?

$$i \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle \xrightarrow{\text{rewrite}} \frac{dx(t)}{dt} = Ax(t) \quad \text{with} \quad x \in \mathbb{C}^d, \quad A = -iH \xrightarrow{\text{solve}} x(t) = e^{At}x(0)$$

Can we go beyond
antihermitian A ?

Key concept

$A = d \times d$ complex matrix



$$U_A = \begin{bmatrix} A/\alpha & \star \\ \star & \star \end{bmatrix}$$

unitary block-encoding of A
with scale factor α



We can nondeterministically
apply A to a quantum state



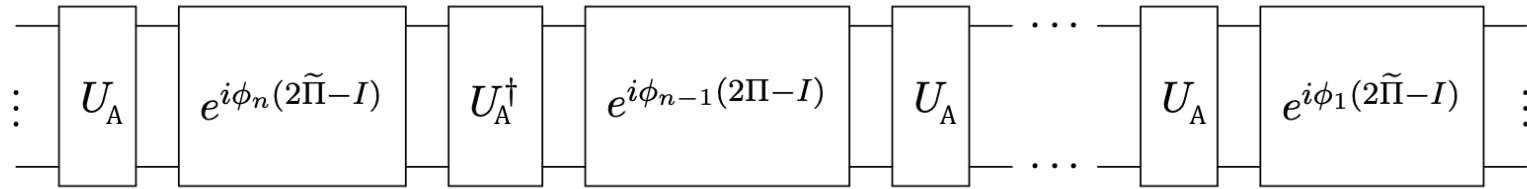
$$U_A(|0\rangle \otimes |\psi\rangle) = |0\rangle \otimes \frac{1}{\alpha} A|\psi\rangle + |\perp\rangle$$

Crucial observation: Some large matrices are easy to encode, requiring only $O(\log d)$ gates.

LINEAR ALGEBRA ON QUANTUM COMPUTERS

Quantum singular value transformation

Central quantum algorithmic result: Given a unitary block-encoding U_A of a general matrix A , one can construct a quantum circuit that applies a polynomial transformation to the singular values of A :



arXiv:1806.01838

This way we can obtain circuits implementing polynomial approximations of, e.g.,

e^{At} - allowing one to evolve a general ODE: $x(t) = e^{At}x(0)$

A^{-1} - allowing one to solve a linear system: $x = A^{-1}b$

Quantum linear solvers

Problem: Find x satisfying $Ax = b$ with dimension d exponentially large and the condition number $\kappa_A := \|A\|\|A^{-1}\|$.

Quantum solution

Assume you can efficiently construct quantum circuits realizing the following unitaries:

1. Unitary block-encoding U_A of A with scale factor α .
2. Unitary state preparation of the normalized state $|b\rangle = b/\|b\|$.

Then using the above circuits $q = O(\kappa_A \log(1/\epsilon))$ times, QLS prepares a quantum state $|\psi\rangle$ that is ϵ -close to a normalized version of the solution x :

$$|\psi\rangle \approx_{\epsilon} \frac{1}{\|x\|} \sum_{n=1}^d x_n |n\rangle$$

Note that the norm $\|x\|$ can also be estimated.

Also, note that such methods can also be used to solve linear ODEs after discretizing time.

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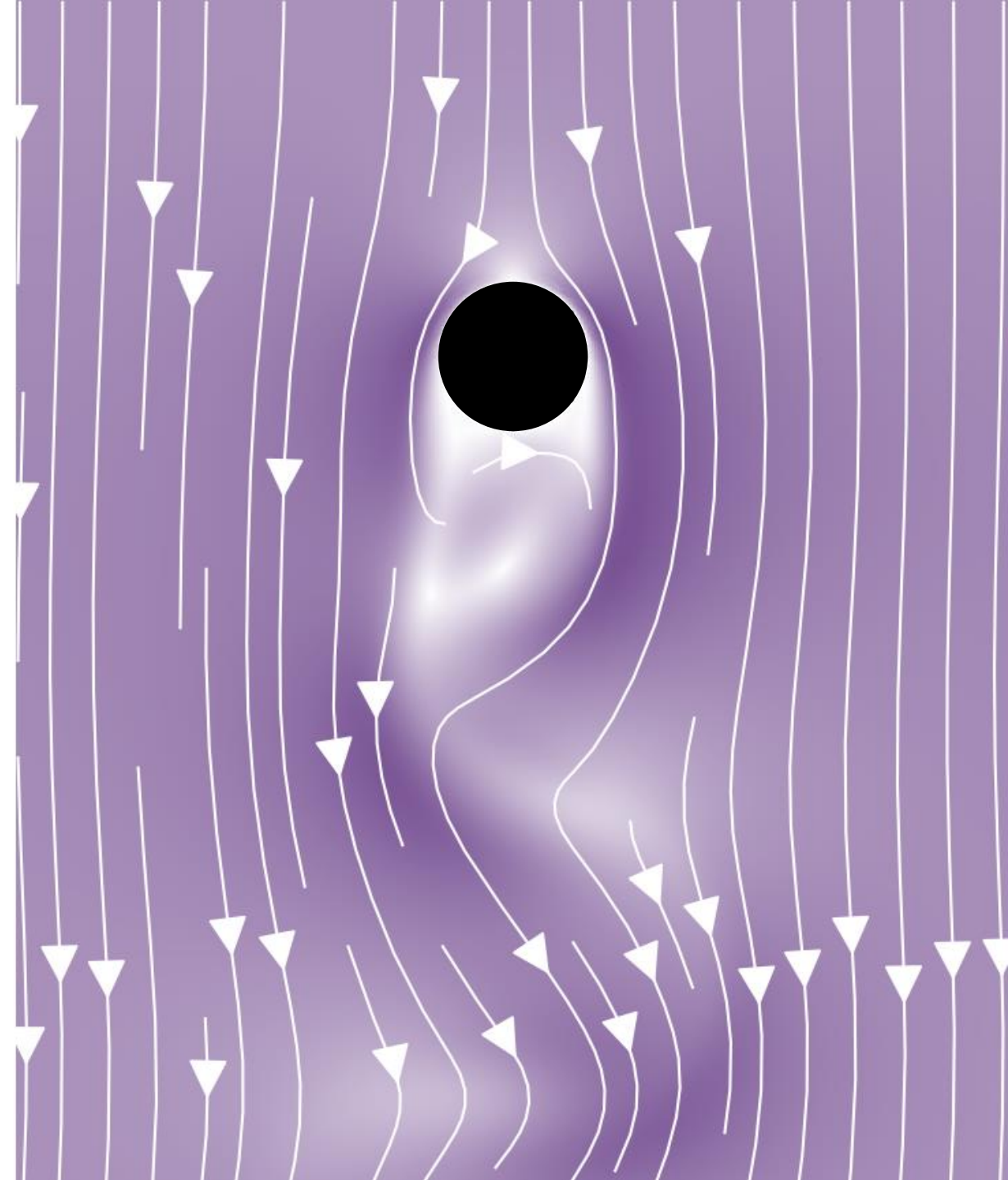
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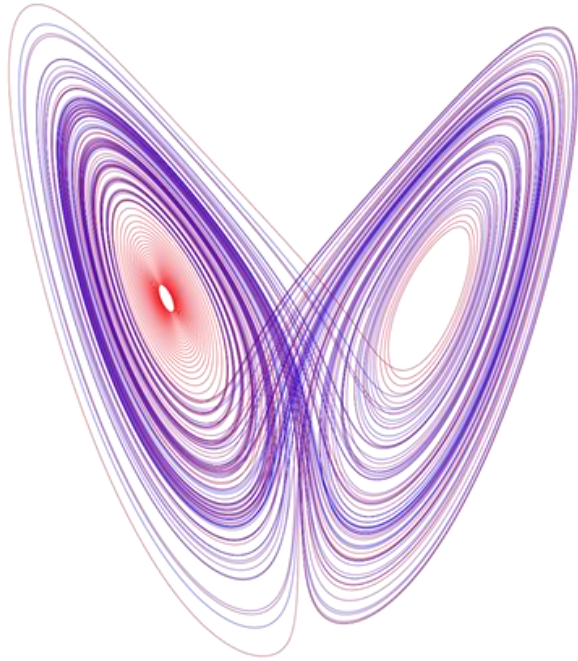


QUANTUM ALGORITHMS FOR NONLINEAR DYNAMICS

The richness of nonlinear dynamics

“The theory of nonlinear dynamics is like the theory of non-elephant animals. “

Stanisław Ulam



QUANTUM ALGORITHMS FOR NONLINEAR DYNAMICS

Carleman embedding

Consider a nonlinear ODE:

$$\frac{d}{dt} x_1(t) = a x_1(t) + b x_1^2(t)$$

Introduce new variable:

$$x_2(t) := x_1^2(t)$$

Get a linear ODE for the old variable:

$$\frac{d}{dt} x_1(t) = a x_1(t) + b x_2(t)$$

But a nonlinear ODE for the new variable

$$\frac{d}{dt} x_2(t) = 2 a x_2(t) + 2 b x_1(t) x_2(t)$$

$$x_3(t) := x_1^3(t)$$

$$\frac{d}{dt} x_2(t) = 2 a x_2(t) + 2 b x_3(t)$$

⋮

⋮

⋮

$$\frac{d}{dt} x_{N_C}(t) = k a x_{N_C}(t) + 2 k b x_{N_C}(t) x_{N_C}(t)$$

Nonlinear ODE with d variables



Linear ODE with $\sim d^{N_C}$ variables

x

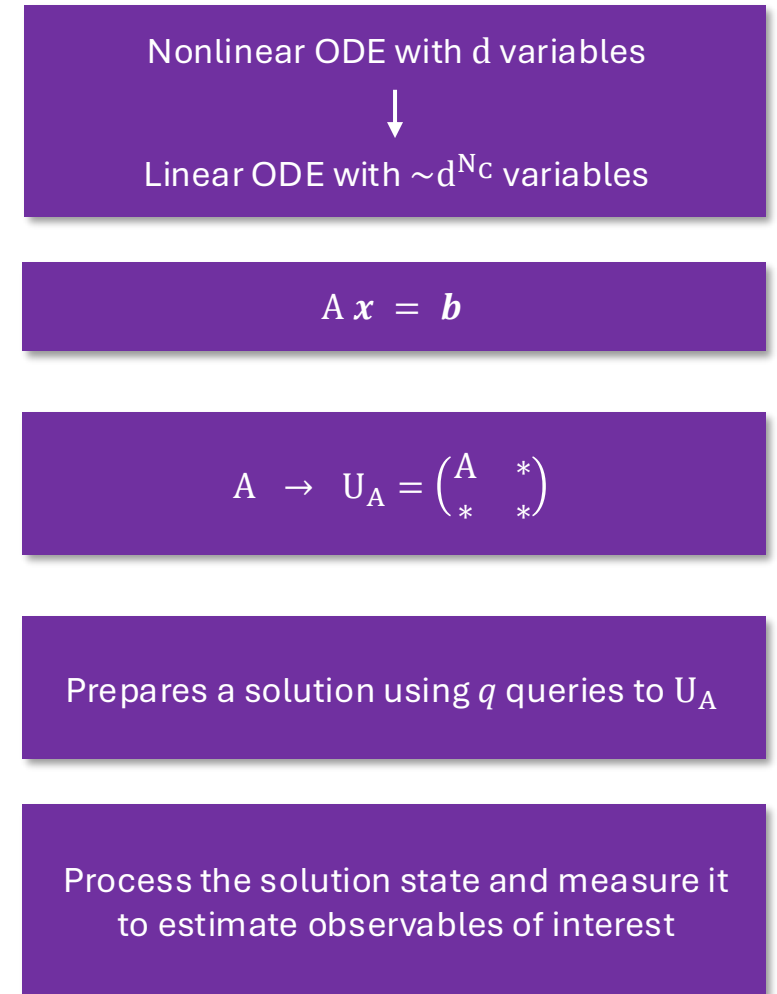
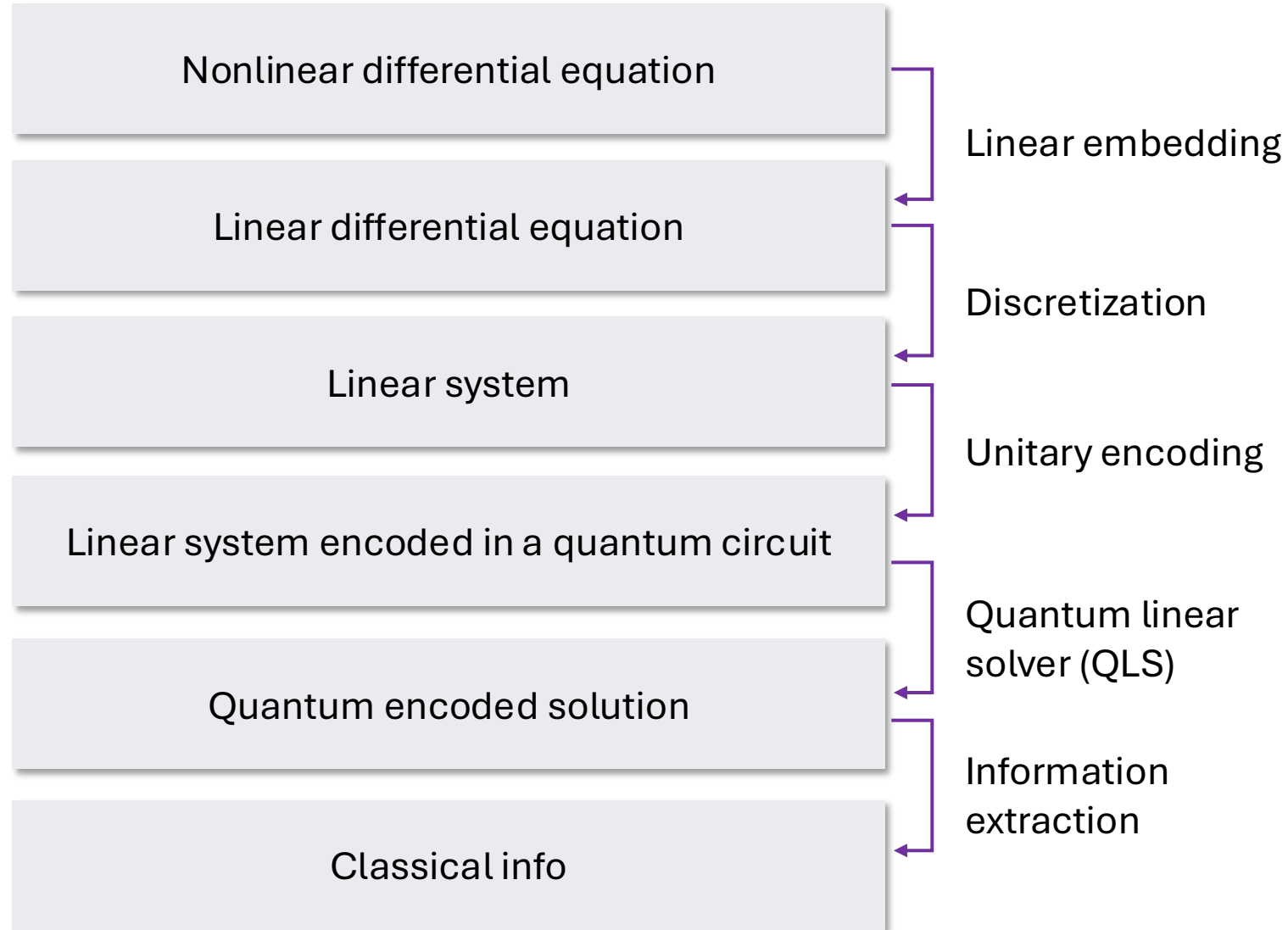


$y = [x, x^{\otimes 2}, \dots, x^{\otimes N_C}]$

Convergence question: how does the truncation error behave with growing N_C ?

QUANTUM ALGORITHMS FOR NONLINEAR DYNAMICS

Algorithmic pipeline



QUANTUM ALGORITHMS FOR NONLINEAR DYNAMICS

Our results

Consider a general quadratic ODE:

$$\frac{d}{dt} x(t) = F_0 + F_1 x(t) + F_2 x^{\otimes 2}(t), \quad x(t) \in \mathbb{C}^N$$

Driving Linear coupling Nonlinearity

Quantum algorithms for general nonlinear dynamics based on the Carleman embedding

David Jennings¹, Kamil Korzekwa¹, Matteo Lostaglio¹, Andrew T Sornborger², Yiğit Subaşı², and Guoming Wang^{*1}

¹PsiQuantum, 700 Hansen Way, Palo Alto, CA 94304, USA

²Computer, Computational, and Statistical Sciences Division, Los Alamos National Laboratory, Los Alamos, New Mexico 87545, USA

[arXiv:2509.07155](https://arxiv.org/abs/2509.07155)

Central Question 1: Carleman convergence

Central Question 2: Algorithmic efficiency

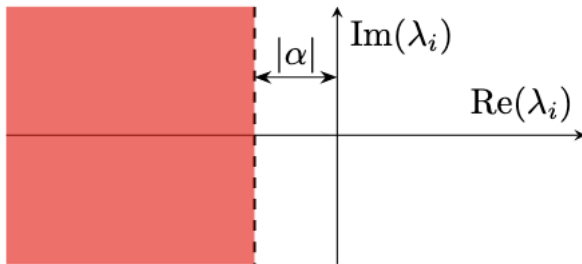
Central Question 3: Exponential advantage in the nonlinear regime

Convergence results

Central Question 1: Carleman convergence

λ_i : Eigenvalues of F_1

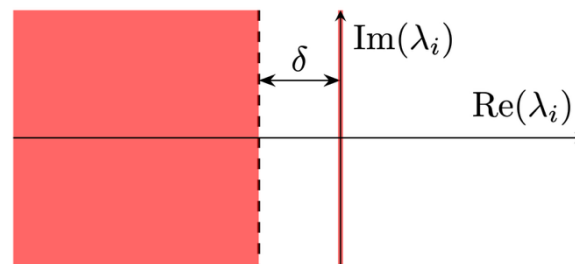
Stable systems



$$R_P < 1$$

$$R_P = \frac{\text{nonlinearity strength}}{\text{stability parameter}}$$

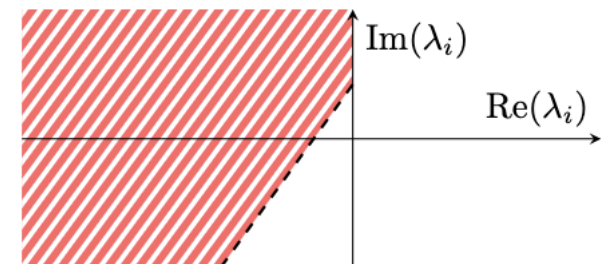
Conservative systems



$$R_\delta < 1$$

$$R_\delta = \frac{\text{nonlinearity strength}}{\text{dissipation parameter}}$$

Nonresonant systems



$$R_\Delta < 1$$

$$R_\Delta = \frac{\text{nonlinearity strength}}{\text{nonresonance parameter}}$$

Algorithmic results

Central Question 2: Algorithmic efficiency

Algorithm 1:

- Assumes F_1 close to a normal matrix.
- Runtime polynomial in T/ϵ .
- Applicable, e.g., to nonlinear Schrödinger equations.

Algorithm 2:

- Applicable only to dissipative systems.
- Assumes access to the Lyapunov matrix for F_1 .
- Runtime logarithmic in T/ϵ .

Central Question 3: Exponential advantage in the nonlinear regime

We introduce a class of nonlinear oscillator problems that are:

- Exponentially hard to simulate classically.
- Efficiently simulated by our quantum algorithm.
- Contain a BQP-complete problem.

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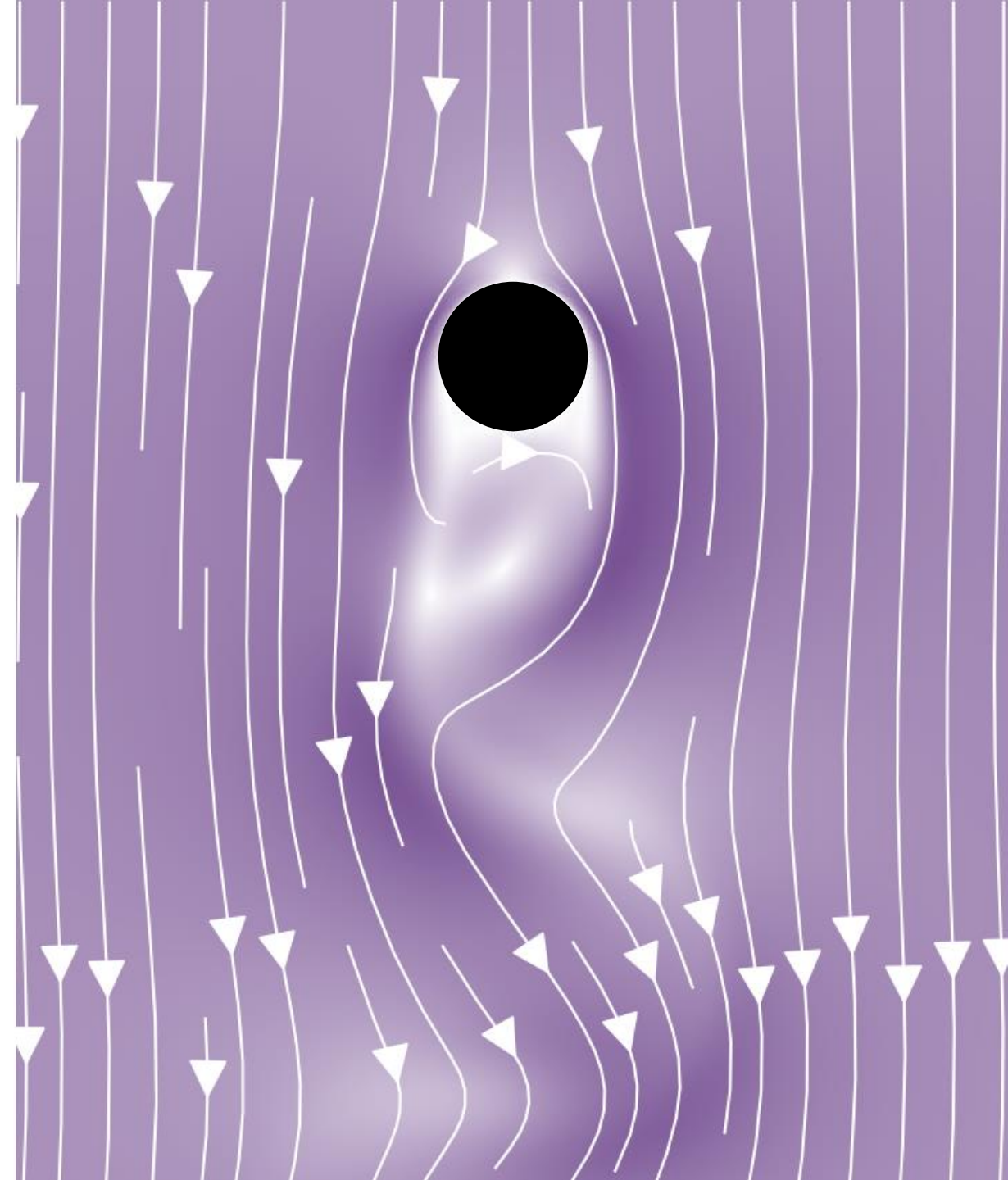
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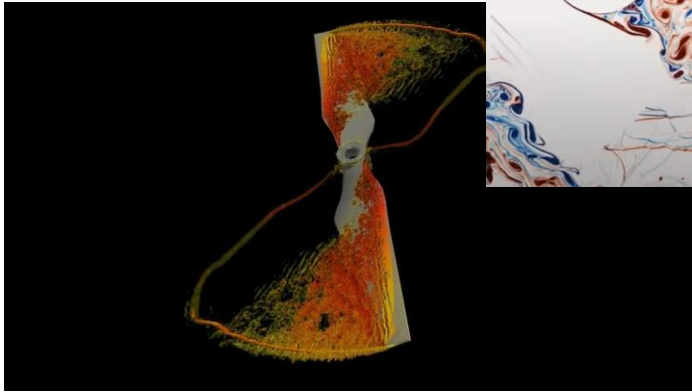
Path for future research



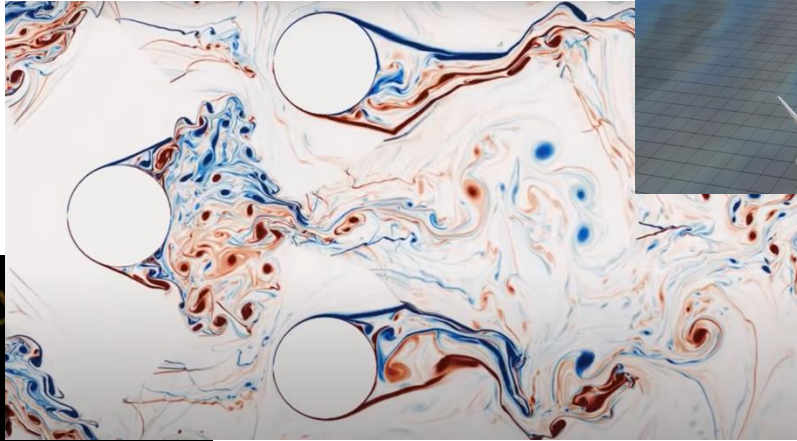
APPLICATION TO FLUID DYNAMICS

Complexity of CFD

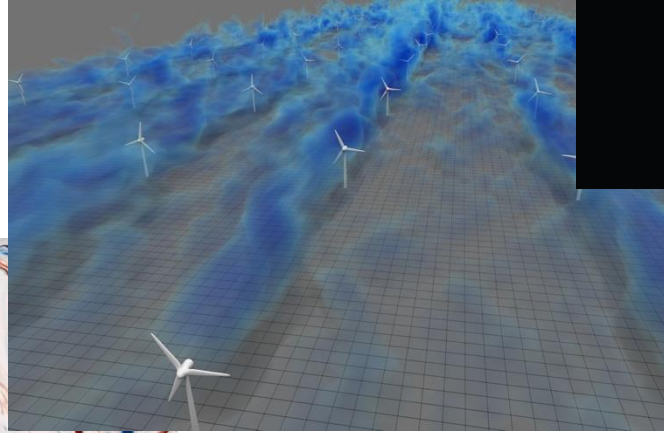
$$\text{Complexity}_C = O(\text{Re}^3)$$



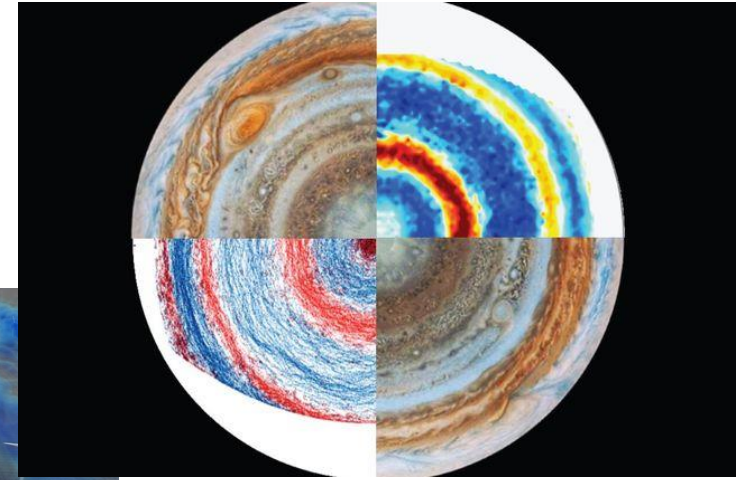
DNS - starting drone rotor $\text{Re} = 1.5 \times 10^4$ (FAU, KTH, KAUST)



DNS - vortex shedding $\text{Re} = 2 \times 10^5$ (John Hopkins)



LES - wind farm $\text{Re} = 10^8$ (John Hopkins)



Experiment – Jupiter's jets
 $\text{Re} \sim 10^{10}$ (UCLA/IRPHE)

$$\text{Complexity}_Q = ?$$

APPLICATION TO FLUID DYNAMICS

Fluid dynamics models

Incompressible Navier-Stokes equations (nonlinearity scales with Re)

Momentum equation $\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p + \nu \Delta \mathbf{u} + \mathbf{F}$

Mass conservation $\nabla \cdot \mathbf{u} = 0$

$\mathbf{u}$ – velocity field

p – pressure field

ν – kinematic viscosity

$\mathbf{F}$ – external force

Discrete-Boltzmann equation (nonlinearity scales with Ma)

$$\partial_t f_m(\mathbf{r}, t) + \mathbf{e}_m \cdot \nabla f_m(\mathbf{r}, t) = \Omega_m(\mathbf{r}, t)$$

$$p = c_s^2 \sum_{m=1}^Q f_m(\mathbf{r}, t)$$

$$\mathbf{u}(\mathbf{r}, t) = \frac{1}{\rho_0} \sum_{m=1}^Q f_m(\mathbf{r}, t) \cdot \mathbf{e}_m$$

$\mathbf{e}_m \cdot \nabla f_m(\mathbf{r}, t)$ – streaming term

$\Omega_m(\mathbf{r}, t)$ – collision term

$f_m(\mathbf{r}, t)$ – discretized density of particles

$\mathbf{e}_m$ – unit velocity in direction $m \in \{1, \dots, Q\}$

$Q = 3/9/27$ – number of unit velocities available in 1D/2D/3D

APPLICATION TO FLUID DYNAMICS

Our results

Central questions

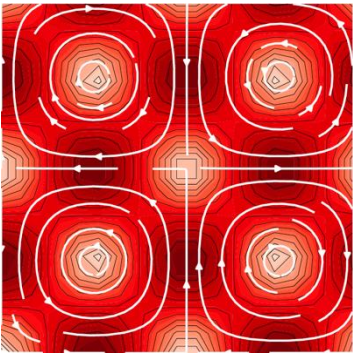
Carleman convergence

How does the embedding error ϵ_C behave with growing N_C ?

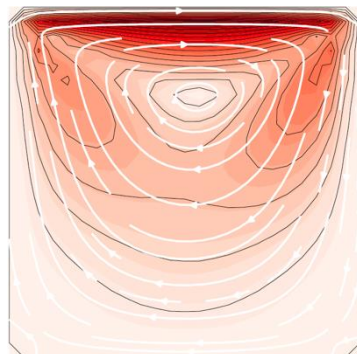
Quantum speedup

How does the quantum computational complexity grow with Re ?

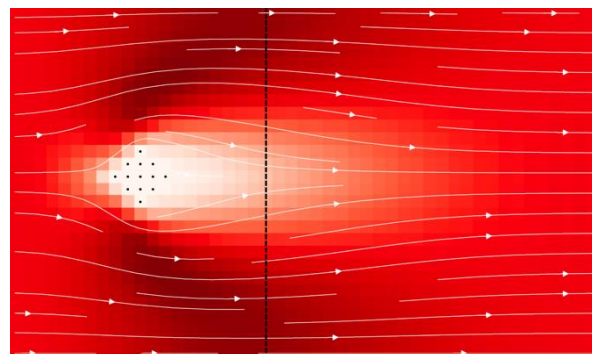
Classical benchmarking on canonical CFD problems



Taylor-Green vortex



Lid-driven cavity flow



Flow past a cylinder

An end-to-end quantum algorithm for nonlinear fluid dynamics with bounded quantum advantage

David Jennings¹, Kamil Korzekwa^{*1}, Matteo Lostaglio¹, Richard Ashworth², Emanuele Marsili², and Stephen Rolston²

¹PsiQuantum, 700 Hansen Way, Palo Alto, CA 94304, USA

²Airbus Operations Ltd, Pegasus House Aerospace Avenue, Filton, Bristol, UK

[arXiv:2512.03758](https://arxiv.org/abs/2512.03758)

Simulating non-trivial incompressible flows with a quantum lattice Boltzmann algorithm

David Jennings^{*}, Kamil Korzekwa^{*†}, Matteo Lostaglio^{*}, and Paul Mannix^{*†}
PsiQuantum, 700 Hansen Way, Palo Alto, CA 94304, USA

Richard Ashworth[‡], Emanuele Marsili[§], and Stephen Rolston[¶]
Airbus Operations Ltd, Pegasus House Aerospace Avenue, Filton, Bristol, UK

[arXiv:2512.05781](https://arxiv.org/abs/2512.05781)

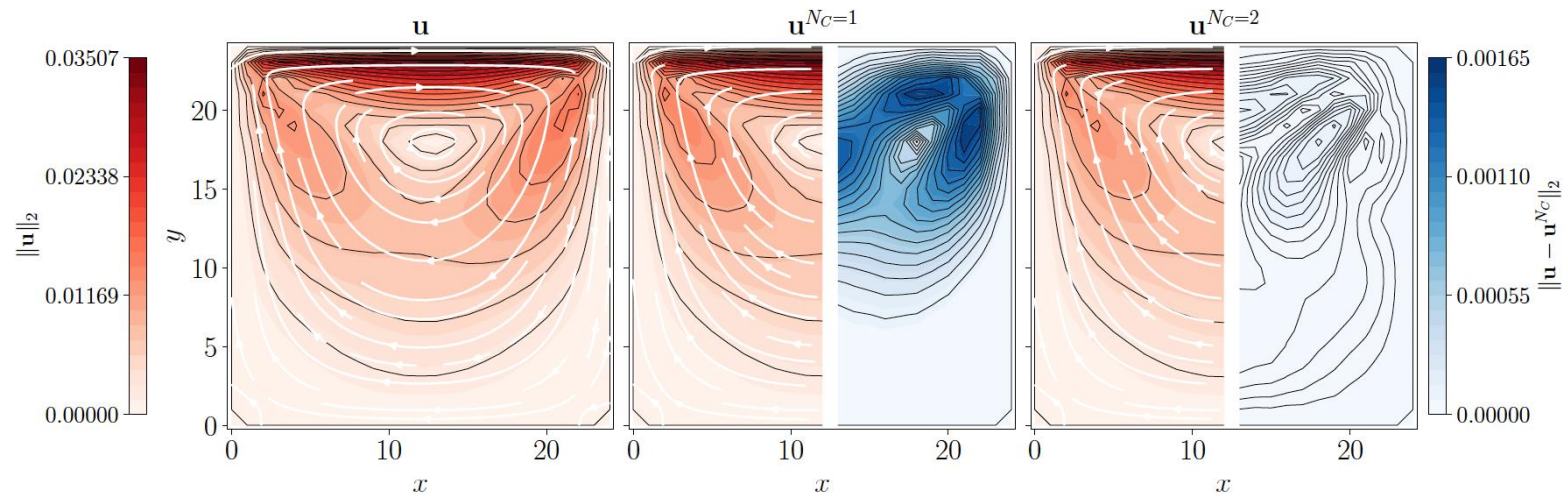
APPLICATION TO FLUID DYNAMICS

Convergence results

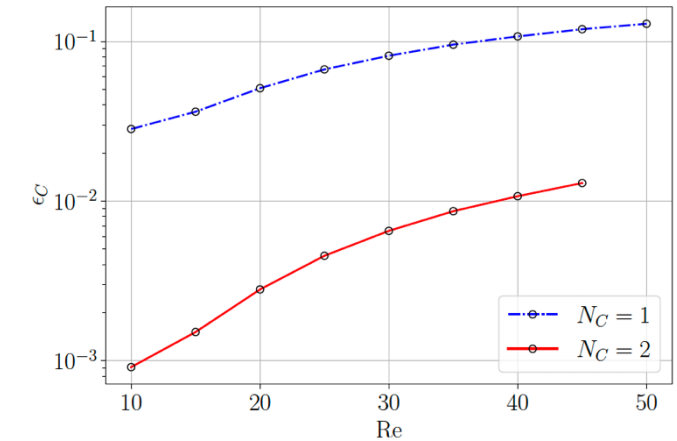
Embedding error measures

$$\epsilon_{rel}(t) := \frac{\|\mathbf{u}^{N_C}(t) - \mathbf{u}(t)\|}{\|\mathbf{u}(t)\|} \quad \epsilon_C(t) := \max_t \epsilon_{rel}(t)$$

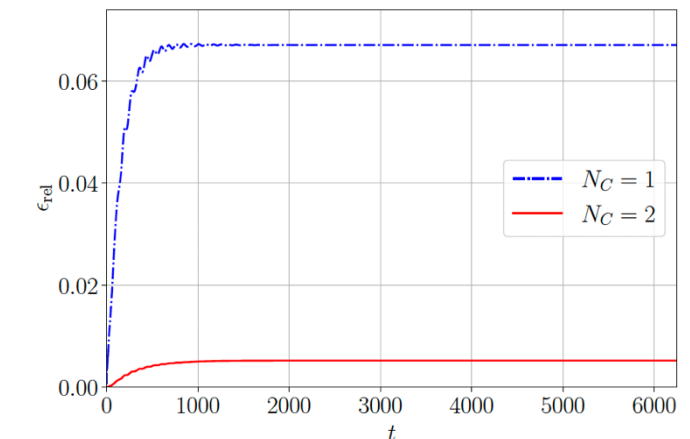
Steady state spatial embedding error for lid-driven cavity flow



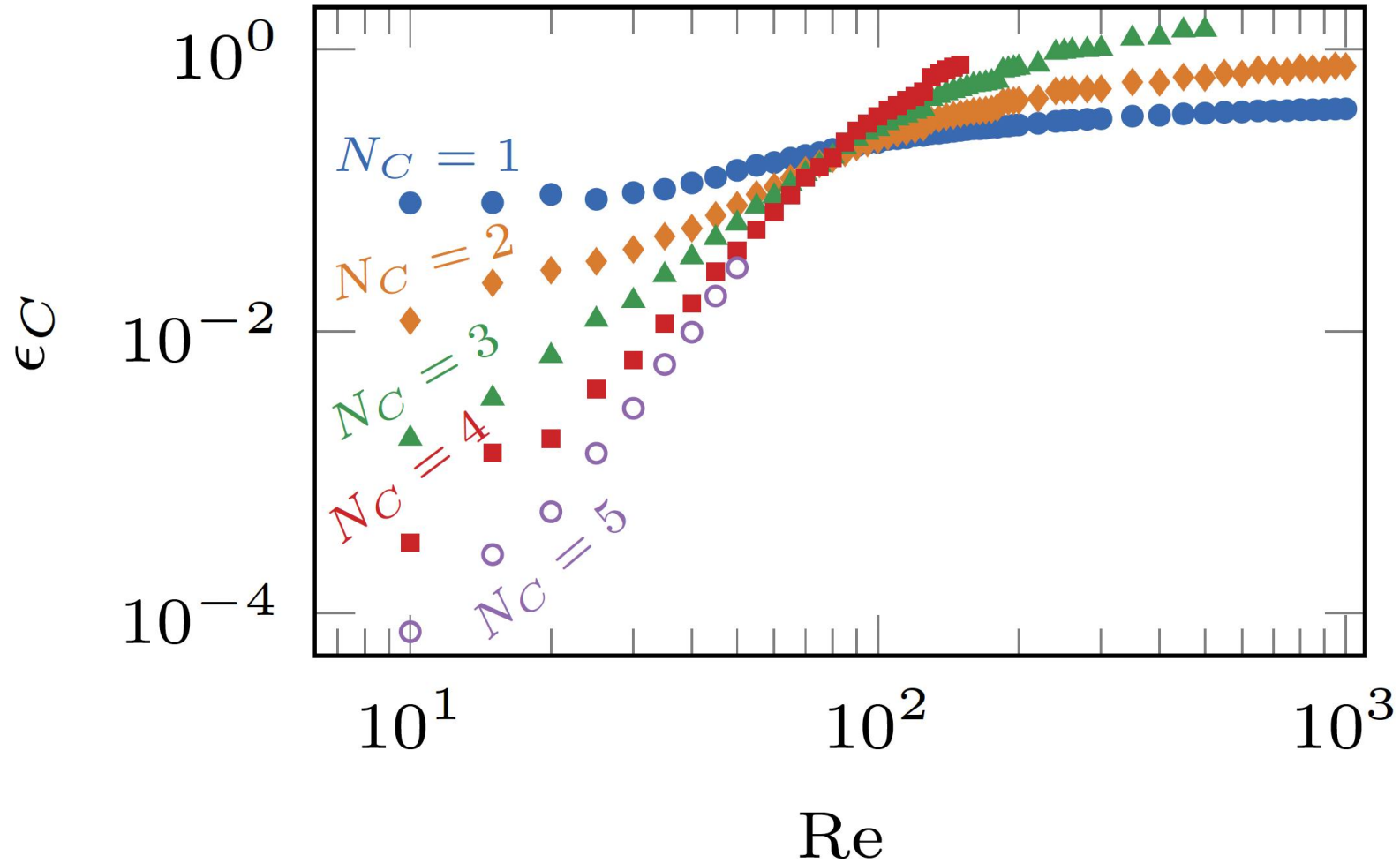
Embedding error convergence



Embedding error boundedness



Convergence results



1D periodic system

Initial velocity field sinusoidal

Quantum speedup results

Quantum complexity contributions

- Initial state preparation
- Block-encoding U_A of the linear system A
- Solving the linear system A
- Information extraction

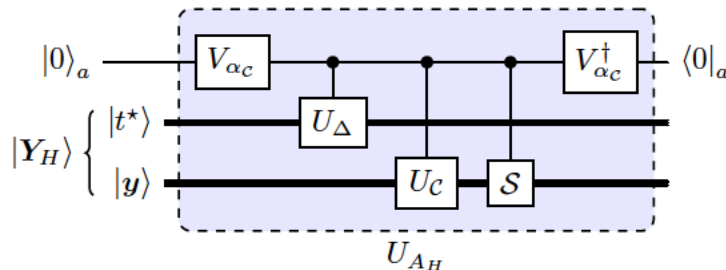
$$\text{Complexity}_Q = g \times q \times q_M$$

Number of gates
in a circuit U_A

Number of circuits
 U_A used by QLS

Measurement
info extraction cost

Compiled the core parts of the circuit to
show that g brings just a constant overhead



For drag force: $q_M = O(Re^{3/8})$

For bulk observables expecting: $q_M = O(1)$

Query complexity q is the
main complexity factor

APPLICATION TO FLUID DYNAMICS

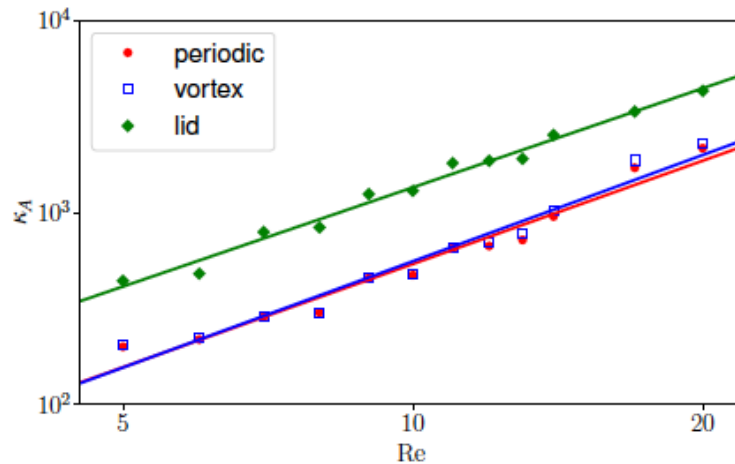
Quantum speedup results

Complexity lower bound

$$\text{Complexity}_Q \geq \text{Re}^{\frac{3}{4}(1+\frac{D}{2})} \quad \text{vs} \quad \text{Complexity}_C = \text{Re}^{\frac{3}{4}(1+D)}$$

At most lower-than-quadratic speedup

Numerical complexity estimate



For 2D problem with $N_C = 2$:

$$\text{Complexity}_Q \simeq \text{Re}^{1.777} \times q_M$$

Modest quantum speedup

Bulk observables: Speedup $\simeq \text{Re}^{0.473}$

Drag force: Speedup $\simeq \text{Re}^{0.098}$

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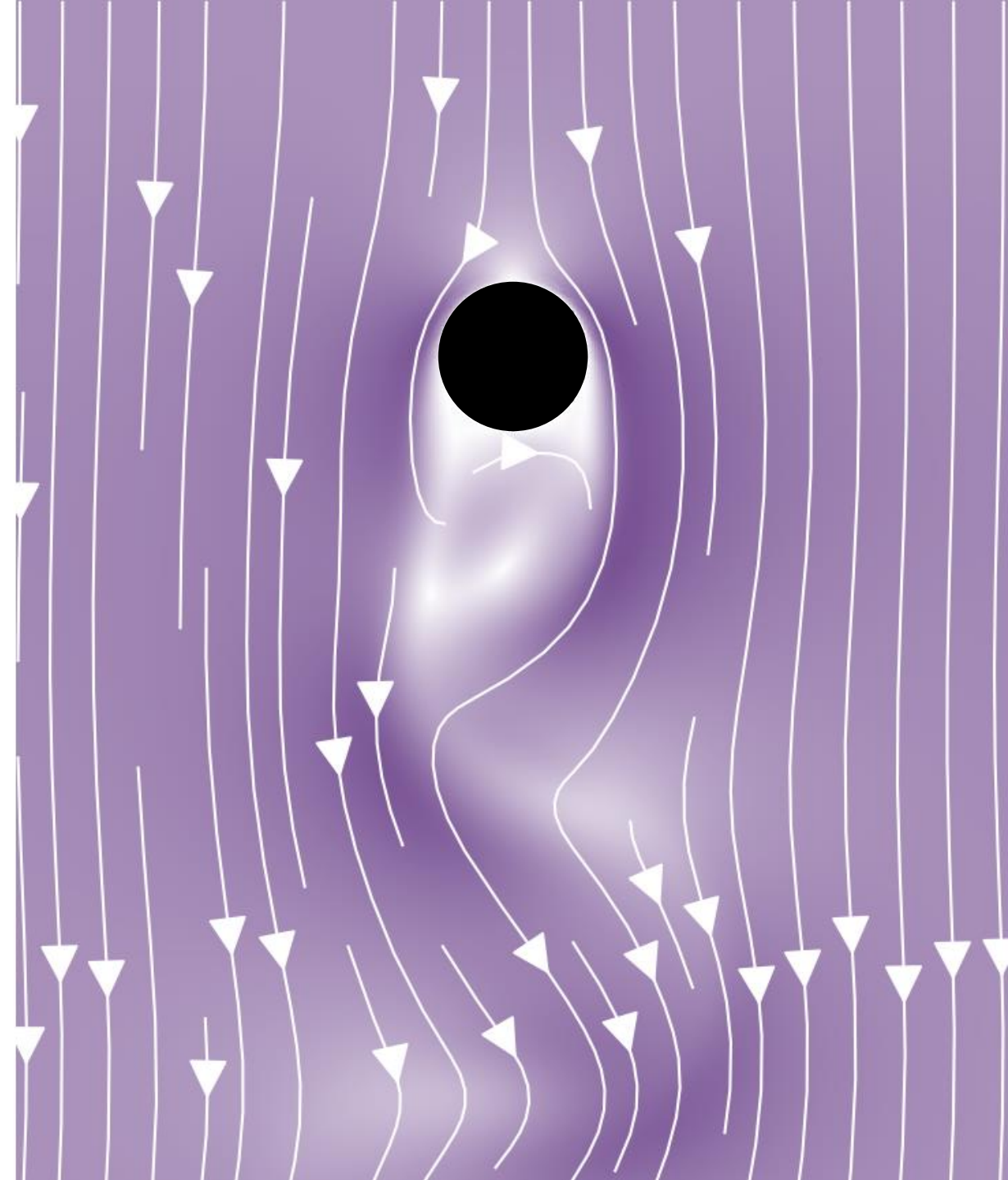
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OUTLOOK

Main takeaways & areas for improvement

Carleman linear embedding converges in some regimes for benchmark problems

Next steps:

- Code optimization & GPU acceleration
- Extend analytical convergence proofs
- Explore alternative linear embeddings*
- Look for high-impact, low-nonlinearity problems

Modest quantum speedup for industrially relevant nonlinear problems is in principle possible

Next steps:

- Investigate preconditioning methods
- Employ novel quantum linear solvers
- Focus on more efficiently extractable observables
- Use spectral methods to reduce complexity

*Check our recent preprint on Quantum Koopman Algorithms: **arXiv:2605.19054**

Thank you!

